

# SSL Frontends

Self-supervised speech representations as detector front ends

2 papers · abstracts sourced directly from arXiv · links open the original paper

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## 1 Automatic Speaker Verification Spoofing and Deepfake Detection Using Wav2vec 2.0 and Data Augmentation

Hemlata Tak, Massimiliano Todisco, Xin Wang, Jee-weon Jung, Junichi Yamagishi, Nicholas Evans

Odyssey 2022 (Speaker and Language Recognition Workshop) · arXiv:2202.12233

■ <https://arxiv.org/abs/2202.12233>

### ABSTRACT

The performance of spoofing countermeasure systems depends fundamentally upon the use of sufficiently representative training data. With this usually being limited, current solutions typically lack generalisation to attacks encountered in the wild. Strategies to improve reliability in the face of uncontrolled, unpredictable attacks are hence needed. We report in this paper our efforts to use self-supervised learning in the form of a wav2vec 2.0 front-end with fine tuning. Despite initial base representations being learned using only bona fide data and no spoofed data, we obtain the lowest equal error rates reported in the literature for both the ASVspoof 2021 Logical Access and Deepfake databases. When combined with data augmentation, these results correspond to an improvement of almost 90% relative to our baseline system.

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## 2 Investigating Self-Supervised Front Ends for Speech Spoofing Countermeasures

Xin Wang, Junichi Yamagishi

Odyssey 2022 (Speaker and Language Recognition Workshop) · arXiv:2111.07725

■ <https://arxiv.org/abs/2111.07725>

### ABSTRACT

Self-supervised speech model is a rapid progressing research topic, and many pre-trained models have been released and used in various down stream tasks. For speech anti-spoofing, most countermeasures (CMs) use signal processing algorithms to extract acoustic features for classification. In this study, we use pre-trained self-supervised speech models as the front end of spoofing CMs. We investigated different back end architectures to be combined with the self-supervised front end, the effectiveness of fine-tuning the front end, and the performance of using different pre-trained self-supervised models. Our findings showed that, when a good pre-trained front end was fine-tuned with either a shallow or a deep neural network-based back end on the ASVspoof 2019 logical access (LA) training set, the resulting CM achieved competitive and generalisable performance across evaluation conditions. [Abstract excerpt; see the arXiv listing for the complete text.]

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